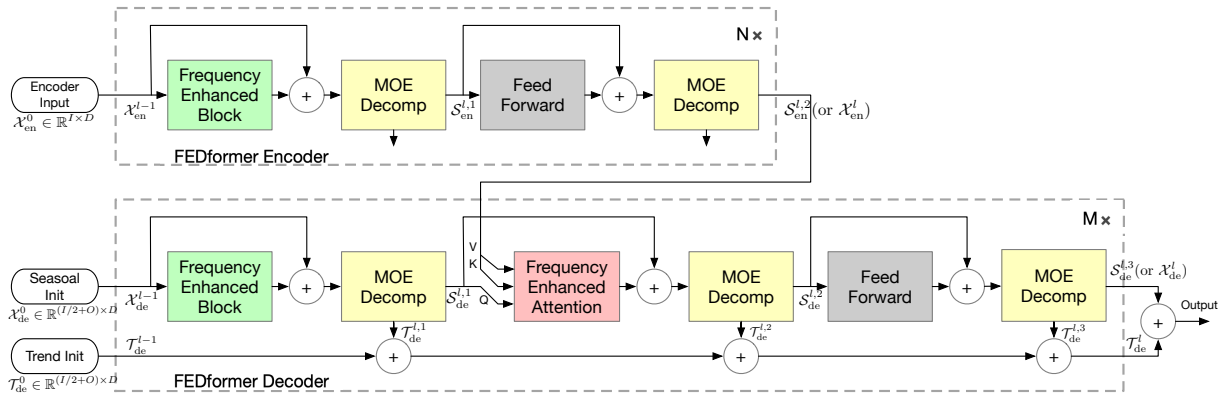

N ×
Encoder Input
$\mathcal{X}_{\text{en}}^{0} \in \mathbb{R}^{I \times D}$
$\mathcal{X}_{\text{en}}^{l-1}$
Frequency Enhanced Block
MOE Decomp
$\mathcal{S}_{\text{en}}^{l,1}$
Feed Forward
MOE Decomp
$\mathcal{S}_{\text{en}}^{l,2}$(or $\mathcal{X}_{\text{en}}^{l}$)
FEDformer Encoder
M ×
Seasoal Init
$\mathcal{X}_{\text{de}}^{0} \in \mathbb{R}^{(I/2+O) \times D}$
$\mathcal{X}_{\text{de}}^{l-1}$
Frequency Enhanced Block
MOE Decomp
$\mathcal{S}_{\text{de}}^{l,1}$
V
K
Q
Frequency Enhanced Attention
MOE Decomp
$\mathcal{S}_{\text{de}}^{l,2}$
Feed Forward
MOE Decomp
$\mathcal{S}_{\text{de}}^{l,3}$(or $\mathcal{X}_{\text{de}}^{l}$)
Output
Trend Init
$\mathcal{T}_{\text{de}}^{0} \in \mathbb{R}^{(I/2+O) \times D}$
$\mathcal{T}_{\text{de}}^{l-1}$
$\mathcal{T}_{\text{de}}^{l,1}$
$\mathcal{T}_{\text{de}}^{l,2}$
$\mathcal{T}_{\text{de}}^{l,3}$
$\mathcal{T}_{\text{de}}^{l}$
FEDformer Decoder